

# Introduction to Internet of Things

Lecture 5

# IoT Levels & Deployment Templates

► An IoT system comprises of the following components:

1. IoT Device
2. Resource ---- software components
3. Controller Service ---- runs on the device and interacts with the web services
4. Database ---- can be either local or in the cloud
5. Web Service ---- link between IoT device, application, database and analysis components
6. Analysis Component --- Analyzing the IoT data
7. Application ---- an interface

# REST vs WebSocket

| REST based service           | WebSocket based service   |
|------------------------------|---------------------------|
| Stateless in nature          | Stateful in nature        |
| Uni-directional              | Bi-directional            |
| Each time new TCP connection | A single TCP connection   |
| Header Overhead              | Minimal frame information |
| Request-Response             | Full duplex               |
| Scalable                     | Not completely scalable   |

In WebSocket services, vertical scaling is easier than horizontal scaling. For REST services, both horizontal (scaling-out) and vertical scaling (scaling-up) solutions are possible.

## IoT – Level 1 Example : Home Automation System



## IoT – Level 2 Example: **Smart Irrigation**



## IoT – Level 3 Example: **Tracking Package Handling**



### **Sensors used**

#### **Accelrometer**

sense movement or vibrations



#### **Gyroscope**

Gives orientation info



**Websocket** service is used because sensor data can be sent in real time.

## IoT – Level 4 Example: Noise Monitoring

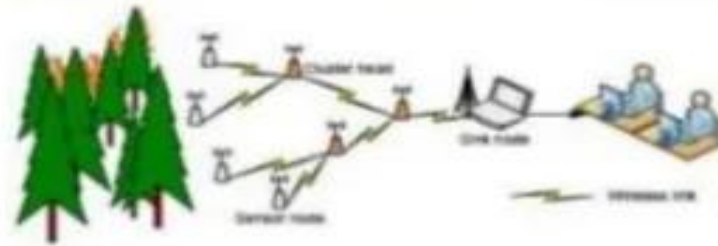
Sound Sensors are used



## IoT – Level 5 Example: **Forest Fire Detection**

Detect forest fire in early stages to take action while the fire is still controllable.

Sensors measure the temperature, smoke, weather, slope of the earth, wind speed, speed of fire spread, flame length



## IoT – Level 6 Example: **Weather Monitoring System**



### **Sensors used**

Wind speed and direction  
Solar radiation  
Temperature (air, water, soil)  
Relative humidity

Precipitation  
Snow depth  
Barometric pressure  
Soil moisture